

## Topic 10: Ethics and Responsible AI Use

### 10.1 Misinformation and Content Authenticity

- Generative AI dramatically lowers the cost of creating convincing but false text, images, audio, and video
- Synthetic media can be used for fraud, political manipulation, defamation, and non-consensual content
- Content credentials (C2PA): a technical standard for attaching verifiable provenance metadata to media ([contentauthenticity.org](https://contentauthenticity.org))
- AI detection tools (GPTZero, Turnitin AI) detect AI-written text -- useful but not infallible
- Best practice: disclose AI use in published content; verify surprising media before sharing

### 10.2 Copyright and Intellectual Property

- Training data: ongoing lawsuits question whether training AI on copyrighted content without permission constitutes infringement
- AI-generated output: US Copyright Office (2024) guidance: purely AI-generated content is generally not copyrightable
- Human-edited AI content may be copyrightable to the extent it reflects genuine human creative choices
- Safe commercial use: Adobe Firefly and Getty AI are trained exclusively on licensed content
- Legal landscape: rapidly evolving -- monitor guidance from copyright offices and courts in your jurisdiction
- EU AI Act (effective 2025): requires labelling of AI-generated content and transparency about training data

### 10.3 Privacy and Data Security

- Public AI tools (ChatGPT free, Claude free): check privacy settings -- inputs may be used to improve models by default
- Never paste personally identifiable information (PII), passwords, API keys, or confidential business data into public AI tools
- Enterprise tiers (ChatGPT Team, Claude for Work, Google Workspace AI): typically include contractual data privacy guarantees
- Local models (Ollama + Llama 3): run entirely on your hardware -- zero data leaves your device ([ollama.com](https://ollama.com))
- GDPR and sector-specific regulations: using AI tools to process personal data triggers compliance obligations

## 10.4 Environmental and Societal Impact

- Energy: training large AI models and running inference at scale consumes significant electricity
  - Water: data centre cooling uses millions of litres of water; major labs are beginning to report water usage
  - Job market: AI is automating many routine cognitive tasks while creating new roles (AI trainers, prompt engineers, AI auditors)
  - Access inequality: frontier AI tools require reliable internet and often a paid subscription -- not universally accessible
  - AI safety research: organisations like Anthropic, DeepMind, and the AI Safety Institute study long-term risks ([anthropic.com/research](https://anthropic.com/research))
  - Responsible use: as a practitioner, consider the downstream impact of what you build and deploy with AI
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